Positivity of the Quartic Solution of a Discrete Painlevé-Type Recurrence

Leslie P. Polzer

Independent Researcher

polzer@fastmail.com

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Abstract. For $a > 0$, we prove the $d = 4$ case of a conjecture of Felder and Hoppe: every leading Gram matrix of their indefinite Hermitian form with density $\exp\bigl(-a|z|^2 + \frac{\mathrm{i}}{4}(z^4 + \bar{z}^4)\bigr)$ is nondegenerate, and all signed norms of the corresponding orthogonal polynomials are positive. Consequently, the solution of the quartic discrete Painlevé-type recurrence arising from the quantum minimal surface $w_2 = w_1^3$, with initial data given by Bessel functions of orders $\pm\frac{1}{4}$, is positive. The proof writes the Bessel amplitude as a positive Laplace integral and uses independent source parameters for z^4 and $\bar{z}^4$ to express each of the four symmetry blocks through a bi-moment matrix with a positive weight. A triangular factorization and Andréief's identity then give every leading block minor as a prescribed sign times an explicit positive multiple integral.

Keywords. discrete Painlevé-type recurrence, indefinite orthogonal polynomials, quartic potential, bi-moment determinant, quantum minimal surface.

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1. Introduction

Discrete Painlevé-type recurrences occur both in matrix regularizations of minimal surfaces and as recurrences for the coefficients of orthogonal polynomials. For the quantum minimal surfaces associated with the holomorphic curves $w_2 = w_1^{d-1}$, recurrences of this type were obtained in [CT99, AHK19, Hop25] (see [FH25, Section 3.1]). The quartic recurrence studied here is written explicitly in [Hop25, equation (38)] and [CDHM25, Section 5]; those works also treat the positive special-function solution in the cubic case $d = 3$.

Felder and Hoppe [FH25] constructed an indefinite Hermitian form for each d and showed that, whenever its leading Gram matrices are nondegenerate, the ratios of its orthogonal-polynomial norms obey

$$V_n + \frac{1}{a^2}\sum_{j=0}^{d-2}\prod_{k=0}^{d-2} V_{n+j-k} = \frac{n+1}{a}. \tag{1.1}$$

For $d = 4$, after setting $v_n = V_n/a$ and $\epsilon = a^{-2}$, the recurrence becomes

$$v_n\,(1 + v_{n-1}v_{n-2} + v_{n+1}v_{n-1} + v_{n+2}v_{n+1}) = (n+1)\epsilon, \qquad n \geq 0, \tag{1.2}$$

where $v_n = 0$ for $n < 0$. The two initial values are

$$v_0 = -\frac{1}{a}\frac{h'(a)}{h(a)}, \qquad v_1 = -\frac{1}{a}\frac{h''(a)}{h'(a)}, \tag{1.3}$$

with

$$\begin{aligned} h(a) = \frac{\pi^2 a}{4}\Bigg[& J_{-1/4}\left(\frac{a^2}{4}\right)^2 - \sqrt{2}\,J_{-1/4}\left(\frac{a^2}{4}\right) J_{1/4}\left(\frac{a^2}{4}\right) \\ & + J_{1/4}\left(\frac{a^2}{4}\right)^2 \Bigg]. \end{aligned} \tag{1.4}$$

This is the function $h(a) = \int_{\mathbb{C}} \rho_a \,\mathrm{d}A$ of [FH25, equation (3.4)] for $d = 4$ (the density ρ_a is recalled in (2.1)), in the Bessel form given in [FH25, Section 4]; Lemma 4.1 below rederives it. Felder and Hoppe proved the required Gram nondegeneracy for $d = 1, 2, 3$ and conjectured it for every d [FH25, Conjecture 3.6 and Theorem 3.7]; the quartic case is the first one not covered by their argument. Their conjecture is stated for $a \geq 0$. We treat $a > 0$, where all planar moments converge absolutely; at $a = 0$ they do not, and the endpoint would require an oscillatory or regularized interpretation, which we do not discuss.

Theorem 1.1. *Let $a > 0$.*

(i) *Every leading Gram matrix of the Felder–Hoppe Hermitian form for $d = 4$ (Section 2) is nondegenerate, and all its signed norms $\mathfrak{h}_n$ are positive. Thus [FH25, Conjecture 3.6] holds for $d = 4$ and $a > 0$.*

(ii) *The initial values (1.3)–(1.4) are well defined and extend uniquely to a real sequence satisfying (1.2) for every $n \geq 0$. This sequence is $v_n = \mathfrak{h}_{n+1}/(a\mathfrak{h}_n)$; in particular, $v_n(a) > 0$ for every $n \geq 0$.*

Uniqueness in (ii) refers to the prescribed initial data; we do not discuss solutions with other initial values.

The proof is an integral sign mechanism that treats all indices at once. The square map $z \mapsto z^2$ and Gaussian integrals turn the quartic planar integral into a positive Laplace integral (Lemma 4.1), which the Dixon–Ferrar formula identifies with (1.4). Independent source parameters u and v for z^4 and $\bar{z}^4$ turn the Gram block of each residue class modulo 4 into the matrix of mixed derivatives $\partial_u^p \partial_v^q$ of a positive kernel that depends only on uv (Lemmas 4.2 and 4.3). Triangular row operations and nonzero diagonal column rescalings reduce these matrices to bi-moment matrices

$$\left[\int_0^\infty s^p \left(\frac{s^2}{1+s^2}\right)^q \mathrm{d}\mu_r(s)\right]_{p,q=0}^{k-1}$$

with positive measures μ_r. Since s and $s^2/(1+s^2)$ are both strictly increasing, Andréief's identity writes their determinants as positive integrals (Lemma 3.1). This fixes the sign

of every leading block minor and gives an explicit positive multiple-integral formula for it (Proposition 4.4). The determinant step is a continuous Cauchy–Binet argument of the kind used for biorthogonal ensembles [Bor99]. To our knowledge, the new ingredient is the two-source representation: differentiation in a common quartic coupling mixes the insertions of z^4 and $\bar{z}^4$, whereas the (p, q) entry of a block requires their powers separately, and independent source derivatives select them. The construction uses two features of $d = 4$: the square map makes the quartic phase quadratic, and one integration by parts makes the kernels of the residue classes 2 and 3 manifestly positive. We do not consider higher d here.

Section 2 recalls the quartic Hermitian form. Section 3 proves the determinant identity for bi-moment matrices. Section 4 derives the four positive kernels and evaluates the block determinants. Section 5 identifies the norm ratios with the solution of (1.2).

2. The quartic Gram form

Fix $a > 0$ and put

$$\rho_a(z) = \exp\left(-a|z|^2 + \frac{\mathrm{i}}{4}(z^4 + \bar{z}^4)\right), \tag{2.1}$$

the density of [FH25, Section 3.1] for $d = 4$. The Gaussian factor makes all polynomial moments absolutely convergent. Let $\eta = \mathrm{e}^{\pi \mathrm{i}/4}$. Since $\eta^4 = -1$,

$$\rho_a(\mathrm{i}z) = \rho_a(z), \qquad \rho_a(\eta z) = \overline{\rho_a(z)}.$$

Decompose the polynomial ring into C_4-weight spaces

$$\mathbb{C}[z] = \bigoplus_{r=0}^{3} \mathcal{P}_r, \qquad \mathcal{P}_r = \{f \in \mathbb{C}[z] : f(\mathrm{i}z) = \mathrm{i}^r f(z)\} = \operatorname{span}\{z^{r+4q} : q \geq 0\}.$$

For $f \in \mathcal{P}_r$, set $(\sigma f)(z) = (-\eta)^{-r} f(\eta z)$, extend σ linearly to $\mathbb{C}[z]$, and define

$$(f, g)_a = \int_{\mathbb{C}} f(z)\overline{\sigma g(z)}\rho_a(z)\,\mathrm{d}A(z), \qquad \mathrm{d}A = \mathrm{d}(\operatorname{Re} z)\,\mathrm{d}(\operatorname{Im} z), \tag{2.2}$$

which is linear in f and conjugate-linear in g. This is the Hermitian form of [FH25, Proposition 3.1] for $d = 4$, with the right inverse $\chi(r) = r$ $(0 \leq r \leq 3)$ chosen there. If $m = r + 4q$ with $0 \leq r \leq 3$, then

$$\sigma z^m = (-\eta)^{-r}\eta^m z^m = (-1)^{r+q} z^m. \tag{2.3}$$

Since σ preserves each $\mathcal{P}_r$ and $\rho_a(\mathrm{i}z) = \rho_a(z)$, the substitution $z \mapsto \mathrm{i}z$ shows that distinct weight spaces are orthogonal. On $\mathcal{P}_r$, in the basis $z^r, z^{r+4}, \ldots$, the Gram matrix is therefore

$$G^{(r)}_{pq} = (z^{r+4p}, z^{r+4q})_a = (-1)^{r+q} M^{(r)}_{pq}, \qquad M^{(r)}_{pq} = \int_{\mathbb{C}} z^{r+4p}\bar{z}^{r+4q}\rho_a(z)\,\mathrm{d}A(z). \tag{2.4}$$

The substitution $z \mapsto \eta z$ and $\rho_a(\eta z) = \overline{\rho_a(z)}$ give $\overline{M^{(r)}_{qp}} = (-1)^{q-p} M^{(r)}_{pq}$, so each $G^{(r)}$ is a Hermitian matrix; with the orthogonality of the weight spaces, this confirms directly that $(\ ,\ )_a$ is Hermitian.

For $k \geq 1$, let

$$\Delta_k^{(r)} = \det[G_{pq}^{(r)}]_{p,q=0}^{k-1}, \qquad \Delta_0^{(r)} = 1. \tag{2.5}$$

Assume for the moment that all $\Delta_k^{(r)}$ are nonzero. After the monomials $1, z, \dots, z^N$ are grouped by residue modulo 4, their Gram matrix is block diagonal, with leading principal submatrices of the $G^{(r)}$ as blocks. Hence every leading Gram matrix $\big[(z^p, z^q)_a\big]_{p,q=0}^{N}$ is nondegenerate, and Gram–Schmidt gives unique monic orthogonal polynomials $P_n(z) = z^n + \cdots$. Orthogonalizing z^n against lower monomials subtracts only components in the weight space of z^n, so $P_n \in \mathcal{P}_{n \bmod 4}$. Following [FH25, Lemma 3.4 and Example 3.5], for $\ell \in \mathbb{Z}$ put

$$\tau(\ell) = \begin{cases} -1, & \ell \equiv 3 \pmod 4, \\ 1, & \text{otherwise}, \end{cases} \qquad \varrho(n) = (-1)^n \prod_{\ell=0}^{n-1} \tau(\ell) = (-1)^{n+\lfloor n/4 \rfloor}. \tag{2.6}$$

The signed norms $\mathfrak{h}_n$ of [FH25, Section 3.1], denoted h_n there, are defined by

$$(P_n, P_n)_a = \varrho(n)\mathfrak{h}_n. \tag{2.7}$$

The matrices $\big[(z^p, z^q)_a\big]_{p,q=0}^{N}$ are exactly the Gram matrices of [FH25, Conjecture 3.6]. Section 4 proves that all $\Delta_k^{(r)}$ are nonzero and that $\mathfrak{h}_n > 0$.

3. A determinant identity for bi-moment matrices

Let $y > 0$, $\alpha \geq 0$, $\beta \geq 0$, and $b = \beta + \frac{1}{2}$. For $w > 0$ and $p, q \geq 0$, define

$$\Phi_{\alpha,\beta}(w; y) = \int_0^\infty \mathrm{e}^{-ys} s^{\alpha-1/2} (1 + ws^2)^{-b} \, \mathrm{d}s, \qquad D_{pq}^{\alpha,\beta} = \partial_u^p \partial_v^q \Phi_{\alpha,\beta}(uv; y)\big|_{u=v=1}. \tag{3.1}$$

Let μ be the measure $\mathrm{d}\mu(s) = \mathrm{e}^{-ys} s^{\alpha-1/2}(1+s^2)^{-b}\,\mathrm{d}s$ on $(0,\infty)$, put $\psi(s) = s^2/(1+s^2)$, and define the bi-moments

$$B_{pq}(y) = \int_0^\infty s^p \psi(s)^q \, \mathrm{d}\mu(s). \tag{3.2}$$

These integrals converge absolutely because $\alpha \geq 0$ and $y > 0$, and they may be differentiated under the integral sign, locally uniformly in $w, y > 0$: the derivative $\partial_w^j (1+ws^2)^{-b} = (-1)^j (b)_j s^{2j} (1+ws^2)^{-b-j}$ has absolute value at most $(b)_j w^{-j}$, and each y-derivative multiplies the integrand by $-s$. Here $(b)_j = b(b+1)\cdots(b+j-1)$ is the rising Pochhammer symbol. In particular, $\Phi_{\alpha,\beta}(uv; y)$ is smooth in (u, v), and $\partial_y^i B_{0q} = (-1)^i B_{iq}$. We write $S_k = k(k-1)/2$.

Lemma 3.1 *(bi-moment determinants).* *For every $k \geq 1$,*

$$(-1)^{S_k} \det\big[D_{pq}^{\alpha,\beta}\big]_{p,q=0}^{k-1} = \Big(\frac{y}{2}\Big)^{S_k} \prod_{q=0}^{k-1} (b)_q \, \det[B_{pq}]_{p,q=0}^{k-1}, \tag{3.3}$$

and

$$\det[B_{pq}]_{p,q=0}^{k-1} \;=\; \frac{1}{k!}\int_{(0,\infty)^k} \prod_{1\le i<j\le k} (s_j - s_i)\big(\psi(s_j)-\psi(s_i)\big) \prod_{j=1}^{k} \mathrm{d}\mu(s_j) \;>\; 0. \tag{3·4}$$

In particular, $\operatorname{sgn}\det[D_{pq}^{\alpha,\beta}]_{p,q=0}^{k-1} \;=\; (-1)^{S_k}$.

Proof. Fix q. Since $\partial_v^q F(uv)|_{v=1} \;=\; u^q F^{(q)}(u)$,

$$\partial_v^q \Phi_{\alpha,\beta}(uv;y)\big|_{v=1} \;=\; (-1)^q (b)_q \int_0^\infty \mathrm{e}^{-ys}\, s^{\alpha-1/2}\, \frac{(us^2)^q}{(1+us^2)^{b+q}}\, \mathrm{d}s.$$

The substitution $s \;=\; t/\sqrt{u}$ turns $(us^2)^q(1+us^2)^{-q}$ into $\psi(t)^q$ and gives

$$\partial_v^q \Phi_{\alpha,\beta}(uv;y)\big|_{v=1} \;=\; (-1)^q (b)_q\, u^{-\gamma} B_{0q}\big(yu^{-1/2}\big), \qquad \gamma \;=\; \frac{\alpha}{2}+\frac{1}{4}. \tag{3·5}$$

By the chain rule there are coefficients $c_{pi}(y)$, depending on γ but not on g, such that

$$\partial_u^p \Big[u^{-\gamma} g\big(yu^{-1/2}\big)\Big]\Big|_{u=1} \;=\; \sum_{i=0}^{p} c_{pi}(y)\, g^{(i)}(y)$$

for every smooth function g on $(0,\infty)$. The derivative $g^{(p)}$ arises only when all p derivatives fall on the inner function $yu^{-1/2}$, so $c_{pp}(y) \;=\; (-y/2)^p$. Applying this to $g \;=\; B_{0q}$ in (3·5) and using $\partial_y^i B_{0q} \;=\; (-1)^i B_{iq}$ gives

$$D_{pq}^{\alpha,\beta} \;=\; (-1)^q (b)_q \sum_{i=0}^{p} (-1)^i c_{pi}(y) B_{iq}.$$

Thus the $k\times k$ truncations satisfy $D \;=\; TB\Lambda$, where $T_{pi} \;=\; (-1)^i c_{pi}(y)$ for $i \;\le\; p$, $T_{pi} \;=\; 0$ for $i \;>\; p$, and $\Lambda \;=\; \operatorname{diag}\big((-1)^q (b)_q\big)_{q=0}^{k-1}$. Since T is lower triangular with $T_{pp} \;=\; (y/2)^p$, we have $\det T \;=\; (y/2)^{S_k}$, while $\det\Lambda \;=\; (-1)^{S_k}\prod_{q=0}^{k-1}(b)_q$. This proves (3·3).

Andréief's identity [And86, For19], applied to the functions s^p and $\psi(s)^q$, all of whose products are μ-integrable, gives

$$\det[B_{pq}]_{p,q=0}^{k-1} \;=\; \frac{1}{k!}\int_{(0,\infty)^k} \det[s_j^p]_{p=0,\dots,k-1}^{j=1,\dots,k}\, \det[\psi(s_j)^q]_{q=0,\dots,k-1}^{j=1,\dots,k} \prod_{j=1}^{k} \mathrm{d}\mu(s_j).$$

Both determinants are Vandermonde determinants, which gives the integrand in (3·4). Since s and ψ are strictly increasing on $(0,\infty)$, this integrand is positive off the diagonals $s_i \;=\; s_j$, which are $\mu^{\otimes k}$-null, and μ has a positive density. Hence $\det B \;>\; 0$. Finally, $(b)_q \;>\; 0$ because $b \;>\; 0$. ∎

4. The four positive kernels

Throughout this section,

$$y \;=\; \frac{a^2}{2}, \qquad C \;=\; \sqrt{\frac{\pi}{2}}.$$

For $R > 0$ and $u, v > 0$, expand the exponential below and integrate termwise; the series converges uniformly in θ. Only the terms with equal powers of $\mathrm{e}^{4\mathrm{i}\theta}$ and $\mathrm{e}^{-4\mathrm{i}\theta}$ survive, so

$$\int_0^{2\pi} \exp\left(\frac{\mathrm{i}R^4}{4}(u\mathrm{e}^{4\mathrm{i}\theta} + v\mathrm{e}^{-4\mathrm{i}\theta})\right) \mathrm{d}\theta \;=\; 2\pi \sum_{m\ge 0} \frac{(-R^8 uv/16)^m}{(m!)^2} \;=\; 2\pi J_0\left(\frac{R^4\sqrt{uv}}{2}\right). \quad (4.1)$$

Lemma 4.1 *(positive source).* *For every $\lambda > 0$,*

$$\int_{\mathbb{C}} \exp\left(-a|z|^2 + \frac{\mathrm{i}\lambda}{4}(z^4 + \bar{z}^4)\right) \mathrm{d}A(z) \;=\; \pi \int_0^\infty \mathrm{e}^{-at} J_0\left(\frac{\lambda t^2}{2}\right) \mathrm{d}t \;=\; C\,\Phi_{0,0}(\lambda^2; y). \quad (4.2)$$

For $\lambda = 1$, these three quantities equal the Bessel combination $h(a)$ of (1.4).

Proof. The integrand on the left is bounded by $\mathrm{e}^{-a|z|^2}$. Polar coordinates $z = R\mathrm{e}^{\mathrm{i}\theta}$, (4.1) with $u = v = \lambda$, and $t = R^2$ give the first equality.

For the second, the map $z \mapsto w = z^2$ is two-to-one from $\mathbb{C}\setminus\{0\}$ onto itself with Jacobian $4|z|^2 = 4|w|$, and $z^4 + \bar{z}^4 = 2(X^2 - Y^2)$ for $w = X + \mathrm{i}Y$. Hence the left-hand side of (4.2) equals

$$\frac{1}{2}\int_{\mathbb{R}^2} \frac{\mathrm{e}^{-a|w|}}{|w|}\, \mathrm{e}^{\mathrm{i}\lambda(X^2 - Y^2)/2}\, \mathrm{d}X\, \mathrm{d}Y.$$

By [OLBC10, equations (10.32.10) and (10.39.2)] with $\nu = -\frac{1}{2}$,

$$\frac{\mathrm{e}^{-a|w|}}{|w|} \;=\; \frac{1}{\sqrt{\pi}} \int_0^\infty \sigma^{-1/2} \exp\left(-\sigma|w|^2 - \frac{a^2}{4\sigma}\right) \mathrm{d}\sigma.$$

The resulting triple integral converges absolutely, because $\int_{\mathbb{R}^2} \mathrm{e}^{-\sigma|w|^2}\,\mathrm{d}X\,\mathrm{d}Y = \pi/\sigma$ and $\int_0^\infty \sigma^{-3/2}\mathrm{e}^{-a^2/(4\sigma)}\,\mathrm{d}\sigma < \infty$. By Fubini's theorem and the Gaussian integrals $\int_{\mathbb{R}} \mathrm{e}^{-(\sigma \mp \mathrm{i}\lambda/2)X^2}\,\mathrm{d}X = \sqrt{\pi}\,(\sigma \mp \mathrm{i}\lambda/2)^{-1/2}$ (principal branch), whose product is $\pi(\sigma^2 + \lambda^2/4)^{-1/2}$, the left-hand side equals

$$\frac{\sqrt{\pi}}{2} \int_0^\infty \sigma^{-1/2}\mathrm{e}^{-a^2/(4\sigma)} \left(\sigma^2 + \frac{\lambda^2}{4}\right)^{-1/2} \mathrm{d}\sigma.$$

The substitution $\sigma = 1/(2s)$, for which $\sigma^2 + \lambda^2/4 = (1 + \lambda^2 s^2)/(4s^2)$, turns this into

$$C \int_0^\infty \frac{\mathrm{e}^{-ys}}{\sqrt{s(1 + \lambda^2 s^2)}}\, \mathrm{d}s \;=\; C\,\Phi_{0,0}(\lambda^2; y).$$

For $\lambda = 1$, the connection formula $J_{-1/4} = (J_{1/4} - Y_{1/4})/\sqrt{2}$ [OLBC10, equation (10.2.3)] reduces (1.4) to

$$h(a) \;=\; \frac{\pi^2 a}{8}\big[J_{1/4}(x)^2 + Y_{1/4}(x)^2\big], \qquad x \;=\; \frac{a^2}{4}.$$

The Dixon–Ferrar formula [DF33] (see also [GR15, entry 6.518]) states that, for $x > 0$ and $|\operatorname{Re}\nu| < \frac{1}{2}$,

$$J_\nu(x)^2 + Y_\nu(x)^2 \;=\; \frac{8\cos(\pi\nu)}{\pi^2} \int_0^\infty K_{2\nu}(2x\sinh t)\, \mathrm{d}t.$$

With $\nu = \frac{1}{4}$, $K_{1/2}(u) = \sqrt{\pi/(2u)}\,\mathrm{e}^{-u}$ [OLBC10, equation (10.39.2)], and $s = \sinh t$, this gives $h(a) = C\,\Phi_{0,0}(1;y)$. The radial representation for $\lambda = 1$ and the Bessel form of h also appear in [FH25, pp. 10–11]. ∎

By Lemma 4.1 with $\lambda = \sqrt{uv}$, the radial generating function

$$\mathcal{Z}(a;u,v) = \pi \int_0^\infty \mathrm{e}^{-at} J_0\left(\frac{t^2\sqrt{uv}}{2}\right) \mathrm{d}t \qquad (u,v > 0) \tag{4.3}$$

satisfies

$$\mathcal{Z}(a;u,v) = C\,\Phi_{0,0}(uv;y), \qquad \mathcal{Z}(a;1,1) = h(a). \tag{4.4}$$

For $u \neq v$, the independently sourced planar density is not absolutely integrable: the factor $\exp\big(\frac{\mathrm{i}}{4}(uz^4 + v\bar{z}^4)\big)$ has modulus $\exp\big(\frac{v-u}{4}\operatorname{Im} z^4\big)$, which defeats the Gaussian factor. We therefore define $\mathcal{Z}$ by the radial integral (4.3); it is the limit of the corresponding disk-truncated planar integrals, and its relation to the planar moments is Lemma 4.3. For $r = 0,1,2,3$, put $\mathcal{Z}_r = (-\partial_a)^r \mathcal{Z}$.

Lemma 4.2 *(four positive kernels).* *Define*

r	A_r	α_r	β_r
0	C	0	0
1	aC	1	0
2	$2C$	1	1
3	$2aC$	2	1

(4.5)

Then $\mathcal{Z}_r(a;u,v) = A_r \Phi_{\alpha_r,\beta_r}(uv;y)$ *for* $r = 0,1,2,3$.

Proof. Write $w = uv$. Since $\partial_a = a\partial_y$ and $\partial_y \Phi_{\alpha,\beta} = -\Phi_{\alpha+1,\beta}$, the cases $r = 0,1$ follow from (4.4). Next, $(-\partial_a)^2 = \partial_y + 2y\partial_y^2$, so $\mathcal{Z}_2 = C\big(2y\Phi_{2,0} - \Phi_{1,0}\big)(w;y)$. Integration by parts, whose boundary terms vanish at 0 and ∞, gives

$$y\int_0^\infty \frac{\mathrm{e}^{-ys}s^{3/2}}{\sqrt{1+ws^2}}\,\mathrm{d}s = \frac{1}{2}\int_0^\infty \frac{\mathrm{e}^{-ys}s^{1/2}}{\sqrt{1+ws^2}}\,\mathrm{d}s + \int_0^\infty \frac{\mathrm{e}^{-ys}s^{1/2}}{(1+ws^2)^{3/2}}\,\mathrm{d}s, \tag{4.6}$$

that is, $y\Phi_{2,0} = \frac{1}{2}\Phi_{1,0} + \Phi_{1,1}$. Hence $\mathcal{Z}_2 = 2C\Phi_{1,1}$, and $\mathcal{Z}_3 = -a\partial_y \mathcal{Z}_2 = 2aC\Phi_{2,1}$. ∎

Lemma 4.3 *(source derivatives and planar moments).* *For* $r \in \{0,1,2,3\}$ *and* $p,q \geq 0$,

$$\partial_u^p \partial_v^q \mathcal{Z}_r(a;u,v)\big|_{u=v=1} = \left(\frac{\mathrm{i}}{4}\right)^{p+q} \int_{\mathbb{C}} z^{r+4p}\bar{z}^{r+4q} \rho_a(z)\,\mathrm{d}A(z). \tag{4.7}$$

Proof. Fix a compact rectangle $Q \subset (0,\infty)^2$ containing $(1,1)$ and a compact interval $I \subset (0,\infty)$ containing a, with left endpoint a_0. Every derivative of J_0 is a finite linear combination of integer-order Bessel functions, which are bounded by 1 on the real axis. By induction on $p+q$,

$$\partial_u^p \partial_v^q J_0\left(\frac{t^2\sqrt{uv}}{2}\right) = \sum_{j=0}^{p+q} a_j(u,v)\,t^{2j} J_0^{(j)}\left(\frac{t^2\sqrt{uv}}{2}\right)$$

with coefficients a_j that are smooth, hence bounded, on Q, and each $-\partial_a$ produces a factor t. Hence, for fixed p, q, r,

$$\left|\partial_u^p \partial_v^q (-\partial_a)^r \left[\mathrm{e}^{-at} J_0\left(\frac{t^2\sqrt{uv}}{2}\right)\right]\right| \le C_{Q,I,p,q,r}\, \mathrm{e}^{-a_0 t}(1+t)^{2p+2q+r} \qquad \big((u,v) \in Q,\ a \in I\big),$$

and dominated convergence allows all these derivatives to pass through the t-integral in (4.3).

For $R > 0$, define the truncations

$$\mathcal{Z}^{[R]}(a;u,v) \;=\; \pi \int_0^{R^2} \mathrm{e}^{-at} J_0\left(\frac{t^2\sqrt{uv}}{2}\right) \mathrm{d}t, \qquad \mathcal{Z}_r^{[R]} \;=\; (-\partial_a)^r \mathcal{Z}^{[R]}.$$

By polar coordinates and (4.1),

$$\mathcal{Z}^{[R]}(a;u,v) \;=\; \int_{|z|\le R} \exp\left(-a|z|^2 + \frac{\mathrm{i}}{4}(uz^4 + v\bar{z}^4)\right) \mathrm{d}A(z).$$

On the compact disk all derivatives in the real parameters u, v, a may be taken under the integral sign. At $u = v = 1$, each u-derivative contributes $\mathrm{i}z^4/4$, each v-derivative contributes $\mathrm{i}\bar{z}^4/4$, and each $-\partial_a$ contributes $|z|^2 = z\bar{z}$. Therefore

$$\partial_u^p \partial_v^q \mathcal{Z}_r^{[R]}(a;u,v)\Big|_{u=v=1} \;=\; \left(\frac{\mathrm{i}}{4}\right)^{p+q} \int_{|z|\le R} z^{r+4p}\bar{z}^{r+4q} \rho_a(z)\, \mathrm{d}A(z).$$

As $R \to \infty$, the left-hand side converges to the left-hand side of (4.7) by the integrable bound above, and the right-hand side converges to the full planar moment because its integrand is bounded by $|z|^{2r+4p+4q}\mathrm{e}^{-a|z|^2}$. ∎

Proposition 4.4 *(block determinants).* *For $r \in \{0,1,2,3\}$ and $k \ge 1$, with the parameters (4.5),*

$$\begin{aligned}(-1)^{rk+S_k}\Delta_k^{(r)} \;=\; & \frac{A_r^k (4a^2)^{S_k}}{k!} \prod_{q=0}^{k-1} \left(\beta_r + \frac{1}{2}\right)_q \\ & \times \int_{(0,\infty)^k} \prod_{j=1}^{k} \frac{\mathrm{e}^{-y s_j}\, s_j^{\alpha_r - 1/2}}{(1+s_j^2)^{\beta_r + k - 1/2}} \prod_{1\le i<j\le k} (s_j - s_i)^2 (s_i + s_j) \prod_{j=1}^{k} \mathrm{d}s_j.\end{aligned} \tag{4.8}$$

The right-hand side is strictly positive, so $\operatorname{sgn}\Delta_k^{(r)} = (-1)^{rk+S_k}$.

Proof. Let $D_{pq}^{(r)} = \partial_u^p \partial_v^q \mathcal{Z}_r(a;u,v)|_{u=v=1}$. Lemma 4.3 and (2.4) give

$$G_{pq}^{(r)} \;=\; (-1)^{r+q} \left(\frac{4}{\mathrm{i}}\right)^{p+q} D_{pq}^{(r)}.$$

Taking $(4/\mathrm{i})^p$ out of row p and $(-1)^{r+q}(4/\mathrm{i})^q$ out of column q, and using $(4/\mathrm{i})^2 = -16$, gives

$$\Delta_k^{(r)} \;=\; \left(\frac{4}{\mathrm{i}}\right)^{2S_k} (-1)^{rk+S_k} \det[D_{pq}^{(r)}]_{p,q=0}^{k-1} \;=\; (-1)^{rk} 16^{S_k} \det[D_{pq}^{(r)}]_{p,q=0}^{k-1}.$$

By Lemma 4.2, $D^{(r)} = A_r D^{\alpha_r,\beta_r}$, so Lemma 3.1 and $16^{S_k}(y/2)^{S_k} = (4a^2)^{S_k}$ give

$$(-1)^{rk+S_k}\Delta_k^{(r)} = A_r^k(4a^2)^{S_k}\prod_{q=0}^{k-1}\left(\beta_r+\frac{1}{2}\right)_q \det[B_{pq}]_{p,q=0}^{k-1}.$$

Finally,

$$\psi(s_j)-\psi(s_i) = \frac{(s_j-s_i)(s_j+s_i)}{(1+s_i^2)(1+s_j^2)},$$

and each factor $1+s_j^2$ occurs $k-1$ times in the product over $i<j$. Substituting this into (3.4) gives (4.8), and its positivity is part of Lemma 3.1. ∎

Corollary 4.5 *(quartic Gram nondegeneracy).* *For every $N \geq 0$, the leading Gram matrix $\big[(z^p,z^q)_a\big]_{p,q=0}^{N}$ is nondegenerate, and the signed norms in (2.7) satisfy*

$$\mathfrak{h}_n > 0 \qquad (n \geq 0).$$

In particular, [FH25, Conjecture 3.6] holds for $d=4$ and $a>0$.

Proof. By Proposition 4.4, every $\Delta_k^{(r)}$ is nonzero, so by Section 2 every leading Gram matrix is nondegenerate and the polynomials P_n exist. Let $n = r+4j$. Replacing z^n by P_n in the basis $z^r, z^{r+4}, \dots, z^{r+4j}$ is a unitriangular change of basis, and orthogonality makes the new Gram matrix $\operatorname{diag}\big([G_{pq}^{(r)}]_{p,q=0}^{j-1}, (P_n,P_n)_a\big)$. Comparing determinants gives

$$(P_n,P_n)_a = \frac{\Delta_{j+1}^{(r)}}{\Delta_j^{(r)}}; \tag{4.9}$$

this Schur-complement identity uses only the nonvanishing of the leading minors, not positive definiteness. By Proposition 4.4, $\operatorname{sgn}(P_n,P_n)_a = (-1)^{r+j}$, and (2.6) gives $\varrho(r+4j) = (-1)^{r+j}$. Hence $\mathfrak{h}_n = \varrho(n)(P_n,P_n)_a > 0$. The last assertion is the identification at the end of Section 2. ∎

Remark 4.6. By (4.9), $\mathfrak{h}_n = |\Delta_{j+1}^{(r)}|/|\Delta_j^{(r)}|$ for $n = r+4j$, and (4.8) writes each determinant as an integral of a positive function. The norms, and the ratios $v_n = \mathfrak{h}_{n+1}/(a\mathfrak{h}_n)$ of Section 5, can therefore be evaluated without the cancellation inherent in the oscillatory moment determinants.

5. Identification with the quartic recurrence

Define

$$V_n = \frac{\mathfrak{h}_{n+1}}{\mathfrak{h}_n}, \qquad v_n = \frac{V_n}{a} \qquad (n \geq 0), \tag{5.1}$$

and $V_j = v_j = 0$ for $j<0$. By Corollary 4.5, $v_n > 0$ for every $n \geq 0$. We derive (1.2), specializing [FH25, Lemma 3.4 and equations (3.1)–(3.3)].

If g has C_4-weight ℓ, then

$$\sigma(zg) = -\tau(\ell) z\,\sigma g. \tag{5.2}$$

For $\ell = 0, 1, 2$ this follows by comparing the twist factors in weights ℓ and $\ell+1$; for $\ell = 3$ the weight returns to 0, and $\eta^4 = -1$ supplies the sign $\tau(3) = -1$.

Let $L = \partial_z + \mathrm{i}z^3$, where $\partial_z = \frac{1}{2}(\partial_{\mathrm{Re}\, z} - \mathrm{i}\partial_{\mathrm{Im}\, z})$. Since $\partial_z \overline{\sigma g(z)} = 0$ and $\partial_z \rho_a(z) = (-a\bar{z} + \mathrm{i}z^3)\rho_a(z)$, integration by parts in (2.2), with no boundary terms because of the Gaussian decay, and (5.2) give, for g of weight ℓ,

$$(Lf, g)_a = a \int_{\mathbb{C}} f\, \bar{z}\, \overline{\sigma g}\, \rho_a \, \mathrm{d}A = -\tau(\ell) a (f, zg)_a. \tag{5.3}$$

The polynomials zP_n and LP_n have weights $n+1$ and $n-1$ modulo 4, degrees $n+1$ and $n+3$, and leading coefficients 1 and i. If $j \le n-7$ and $j \equiv n+1 \pmod 4$, then $\deg LP_j < n$, so (5.3) gives $0 = (LP_j, P_n)_a = -\tau(n) a (P_j, zP_n)_a$. If $j \le n-5$ and $j \equiv n-1 \pmod 4$, then $\deg zP_j < n$, so $(LP_n, P_j)_a = -\tau(j) a (P_n, zP_j)_a = 0$. Hence

$$zP_n = P_{n+1} + c_n P_{n-3}, \qquad LP_n = \mathrm{i}P_{n+3} + d_n P_{n-1}, \tag{5.4}$$

where terms with negative indices are omitted; equivalently, $c_0 = c_1 = c_2 = 0$, and c_n for $n \ge 3$ and d_n for $n \ge 1$ are the coefficients displayed. Among any four consecutive integers exactly one is congruent to 3 modulo 4, so (2.6) gives

$$\varrho(n) = \tau(n)\varrho(n-3), \qquad \varrho(n) = -\tau(n-1)\varrho(n-1).$$

Pairing (5.4) with P_{n-3} (for $n \ge 3$) and P_{n-1} (for $n \ge 1$), and using (5.3) and Hermitian symmetry, we obtain

$$\begin{aligned} \mathrm{i}\varrho(n)\mathfrak{h}_n &= (LP_{n-3}, P_n)_a = -\tau(n) a (P_{n-3}, zP_n)_a, \\ c_n &= \frac{(zP_n, P_{n-3})_a}{\varrho(n-3)\mathfrak{h}_{n-3}} = \frac{\mathrm{i}}{a}\frac{\mathfrak{h}_n}{\mathfrak{h}_{n-3}}, \\ d_n &= \frac{(LP_n, P_{n-1})_a}{\varrho(n-1)\mathfrak{h}_{n-1}} = -\tau(n-1) a \frac{(P_n, zP_{n-1})_a}{\varrho(n-1)\mathfrak{h}_{n-1}} = a\frac{\mathfrak{h}_n}{\mathfrak{h}_{n-1}}; \end{aligned}$$

the first line holds because the leading term of LP_{n-3} is $\mathrm{i}P_n$, and the last because that of zP_{n-1} is P_n. Thus $c_m = \frac{\mathrm{i}}{a} V_{m-3} V_{m-2} V_{m-1}$ for every $m \ge 0$, including $c_m = 0$ for $m < 3$.

Applying the first relation in (5.4) three times shows that the coefficient of P_{n-1} in $z^3 P_n$ is

$$c_n + c_{n+1} + c_{n+2} = \frac{\mathrm{i}}{a} V_{n-1} \left(V_{n-2} V_{n-3} + V_n V_{n-2} + V_{n+1} V_n \right).$$

Let $n \ge 1$. The coefficient of P_{n-1} in $\partial_z P_n$ is n, since $\partial_z P_n = n z^{n-1} + \cdots$ lies in the weight space of P_{n-1}. Comparing coefficients of P_{n-1} in $LP_n = \partial_z P_n + \mathrm{i}z^3 P_n$ and using $d_n = a V_{n-1}$, we obtain, with $m = n-1 \ge 0$,

$$V_m \left[1 + \frac{1}{a^2} \left(V_{m-1} V_{m-2} + V_{m+1} V_{m-1} + V_{m+2} V_{m+1} \right) \right] = \frac{m+1}{a}. \tag{5.5}$$

This is the case $d = 4$ of (1.1), and with $V_m = a v_m$ it is (1.2).

Finally, $P_0 = 1$, $P_1 = z$, and $P_2 = z^2$. By Lemma 4.1, $h(a) = \int_{\mathbb{C}} \rho_a \, \mathrm{d}A$, and differentiation under this absolutely convergent integral gives

$$h'(a) = -\int_{\mathbb{C}} |z|^2 \rho_a(z) \, \mathrm{d}A(z), \qquad h''(a) = \int_{\mathbb{C}} |z|^4 \rho_a(z) \, \mathrm{d}A(z).$$

With (2.3) and $\varrho(0) = -\varrho(1) = \varrho(2) = 1$, this gives

$$\mathfrak{h}_0 = h, \qquad \mathfrak{h}_1 = -h', \qquad \mathfrak{h}_2 = h''.$$

Hence $h(a) > 0$ and $h'(a) < 0$, so (1.3) is well defined, and (5.1) reproduces it. Thus (v_n) extends the initial values and satisfies (1.2) for every $n \geq 0$. If $(\widetilde{v}_n)$ is another real sequence with these properties and $\widetilde{v}_j = v_j$ for $j \leq m+1$, then (1.2) at index m is linear in $\widetilde{v}_{m+2}$ with coefficient $\widetilde{v}_m \widetilde{v}_{m+1} = v_m v_{m+1} > 0$, so $\widetilde{v}_{m+2} = v_{m+2}$. Induction proves uniqueness. This proves Theorem 1.1: part (i) is Corollary 4.5, and part (ii) has just been shown.

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